Time as a Visual Dimension

Workshop 6 — Module 1

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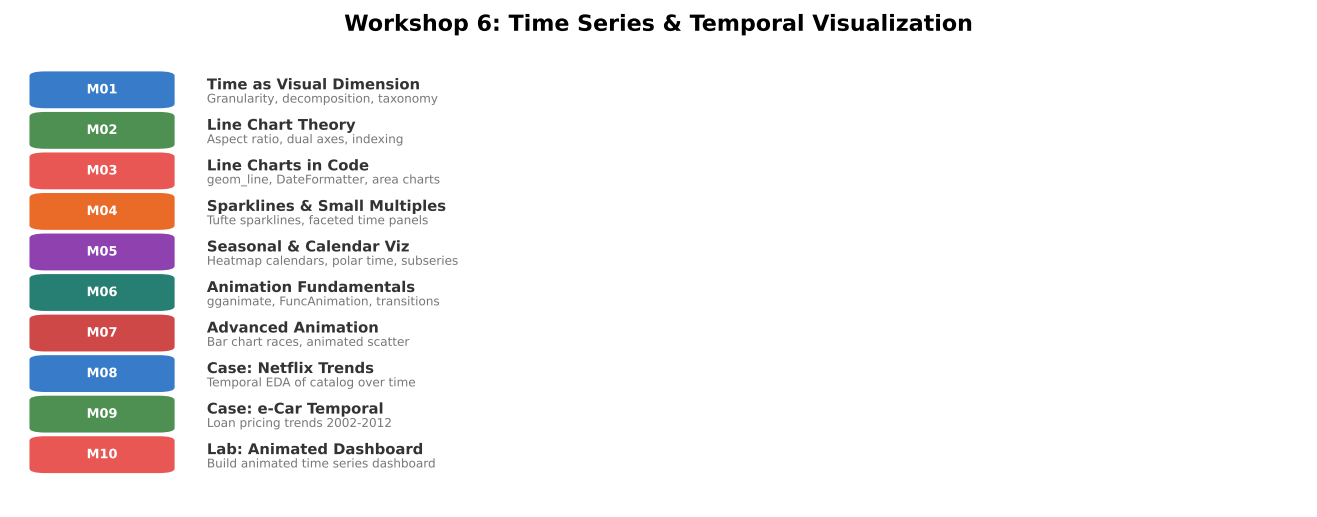
Data Visualization for Data Scientists
UNYT Tirana

Workshop 6 Overview



Module Roadmap

1. What makes time special as a visual dimension
2. Temporal granularity: daily, weekly, monthly, yearly
3. Time series decomposition: trend + seasonal + residual
4. The temporal chart taxonomy: five core types
5. Time-axis pitfalls: spacing, truncation, missing dates

Learning Outcome
You will identify the appropriate temporal granularity for a dataset, decompose a time series into trend, seasonal, and residual components, and choose the right temporal chart type based on the data structure.

Why Time Is Special

Time as the Organising Axis

Time is unique among variables:

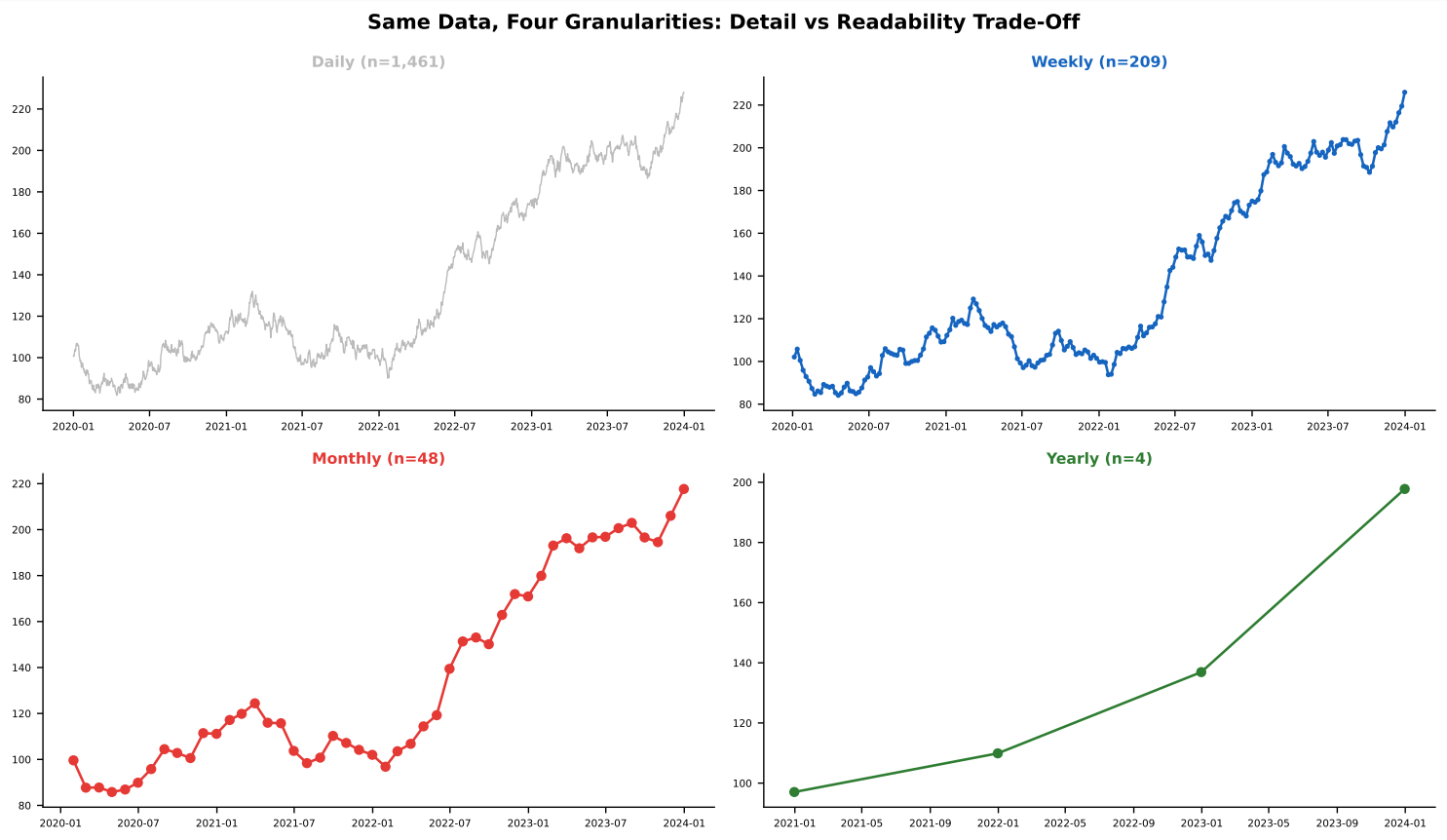
- It is **ordered** and **continuous** (unlike categories)
- It has **natural units** (second, hour, day, week, month, year)
- It enables **causal reasoning**: before $\rightarrow$ after
- It has **hierarchical granularity**: zoom in/out

Because of these properties, temporal data demands specialised chart types (line, area, step) and specialised axis formatting (date breaks, labels, proper spacing). A scatter plot of date vs value "works" but wastes time's natural structure.

Pro-Tip
The x-axis of a time series chart is **not a categorical axis**. It must use a true date/time scale so that gaps between observations reflect real elapsed time. Never convert dates to strings and plot as categories.

Temporal Granularity

Same Data, Four Granularities



Choosing Granularity

Granularity	Best For	Risk
Daily	Short spans (<3 months), event detection	Noisy
Weekly	Medium spans (3–12 months), smooths weekday effects	Loses
Monthly	Annual comparisons, seasonal analysis	Loses
Quarterly	Financial reporting, macro trends	Very
Yearly	Multi-decade trends, cross-country comparison	Loses

Rule: start at the **finest available granularity**, then aggregate up. You can always smooth daily $\rightarrow$ monthly, but you cannot recover daily detail from monthly data.

Resampling in Code

R

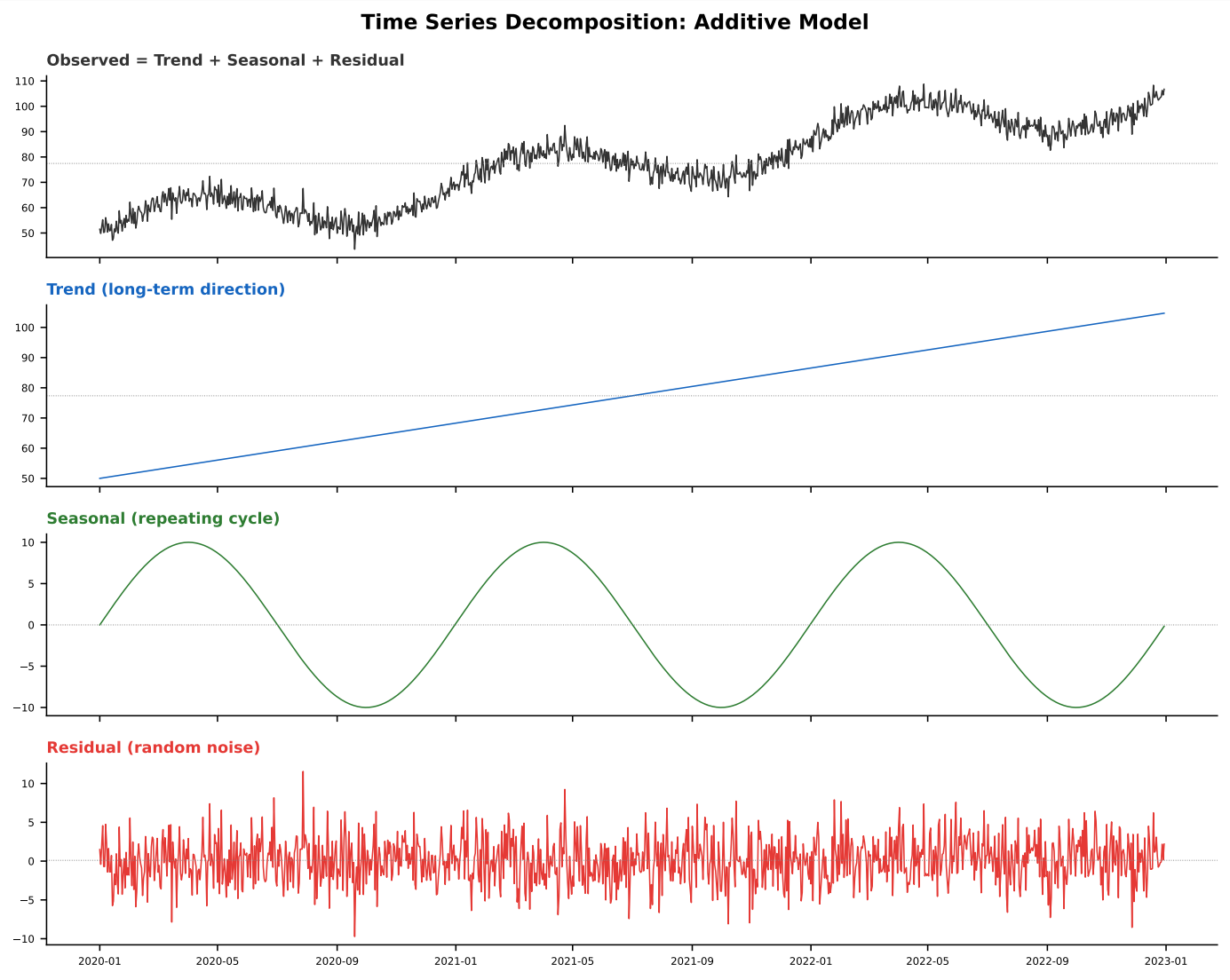
```r
library(lubridate); library(dplyr
    )

daily <- read_csv("data.csv") |>
  mutate(date = ymd(date))

# Weekly mean
weekly <- daily |>
  mutate(week = floor_date(date,
    "week")) |>
  group_by(week) |>
  summarise(value = mean(value))

# Monthly
monthly <- daily |>
  mutate(month = floor_date(date,
    "month")) |>
  group_by(month) |>
  summarise(value = mean(value))
```

Python

```python
df["date"] = pd.to_datetime(df["
    date"])
df = df.set_index("date")

# Weekly mean
weekly = df.resample("W").mean()

# Monthly
monthly = df.resample("ME").mean
    ()

# Quarterly
quarterly = df.resample("QE").
    mean()

# Custom: 2-week rolling average
rolling_14 = df.rolling("14D").
    mean()
```

Time Series Decomposition

Trend + Seasonal + Residual



Two Decomposition Models

Model	Formula	When to Use
Additive	$Y = T + S + R$	Seasonal amplitude is constant (temperature)
Multiplicative	$Y = T \times S \times R$	Seasonal amplitude grows with trend (retail sales)

Pro-Tip
If the "peaks and valleys" get taller as the trend rises, use multiplicative. If they stay the same height regardless of the level, use additive. Quick test: plot the data — if the envelope widens over time, it is multiplicative.

Decomposition in Code

R

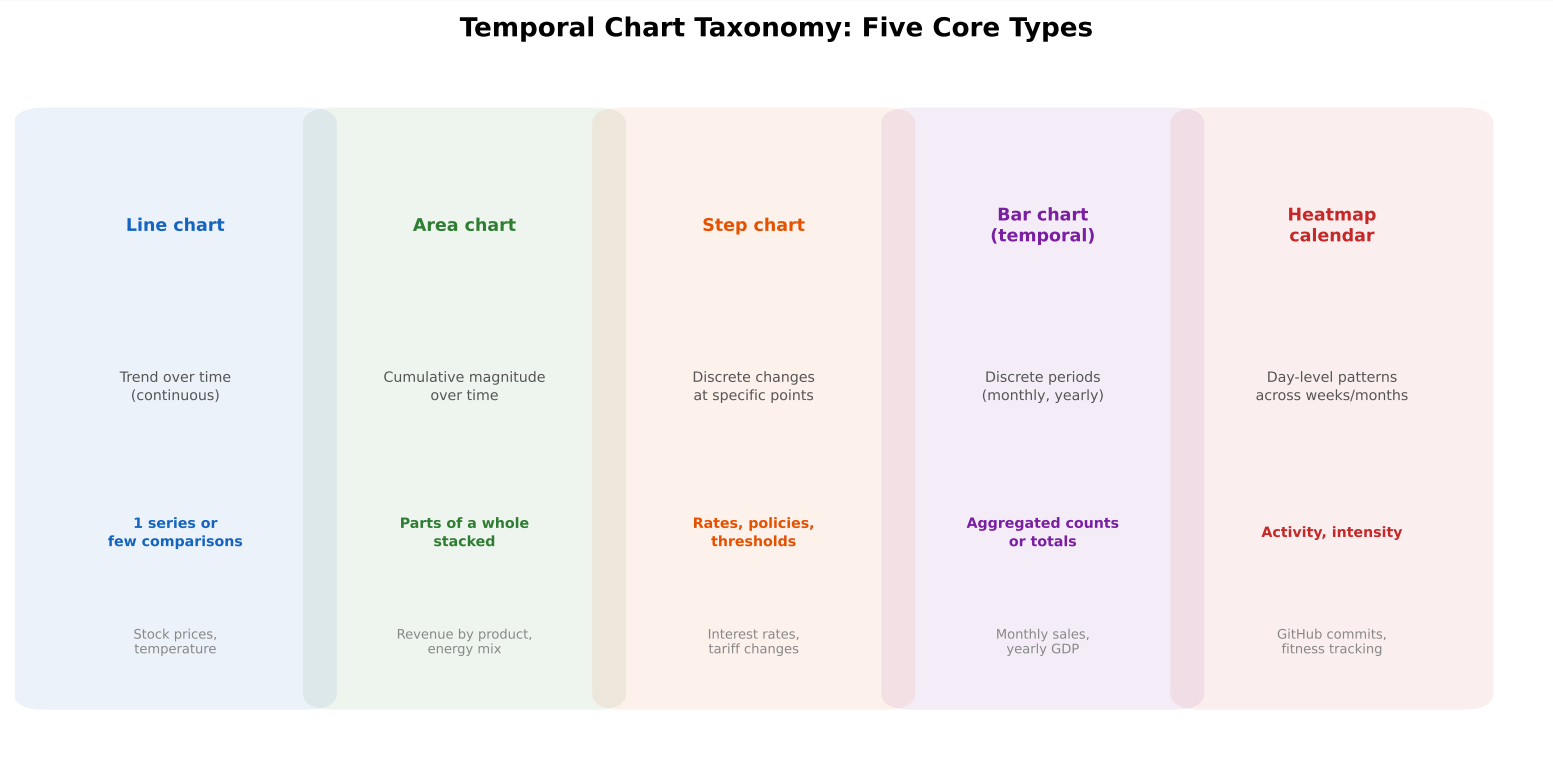
```r
# Base R
ts_data <- ts(daily$value,
  frequency = 365)
decomp <- decompose(ts_data,
  type = "additive")
plot(decomp)

# STL (robust, flexible)
stl_fit <- stl(ts_data,
  s.window = "periodic")
plot(stl_fit)

# Extract components
trend    <- stl_fit$time.series[,
    "trend"]
seasonal <- stl_fit$time.series[,
    "seasonal"]
residual <- stl_fit$time.series[,
    "remainder"]
```

Python

```python
from statsmodels.tsa.seasonal
    import (
    seasonal_decompose, STL)

# Classical decomposition
result = seasonal_decompose(
    df["value"],
    model="additive",
    period=365)
result.plot()

# STL (robust)
stl = STL(df["value"], period
    =365)
res = stl.fit()
res.plot()

# Extract
trend    = res.trend
seasonal = res.seasonal
residual = res.resid
```

Temporal Chart Taxonomy

Five Core Temporal Chart Types

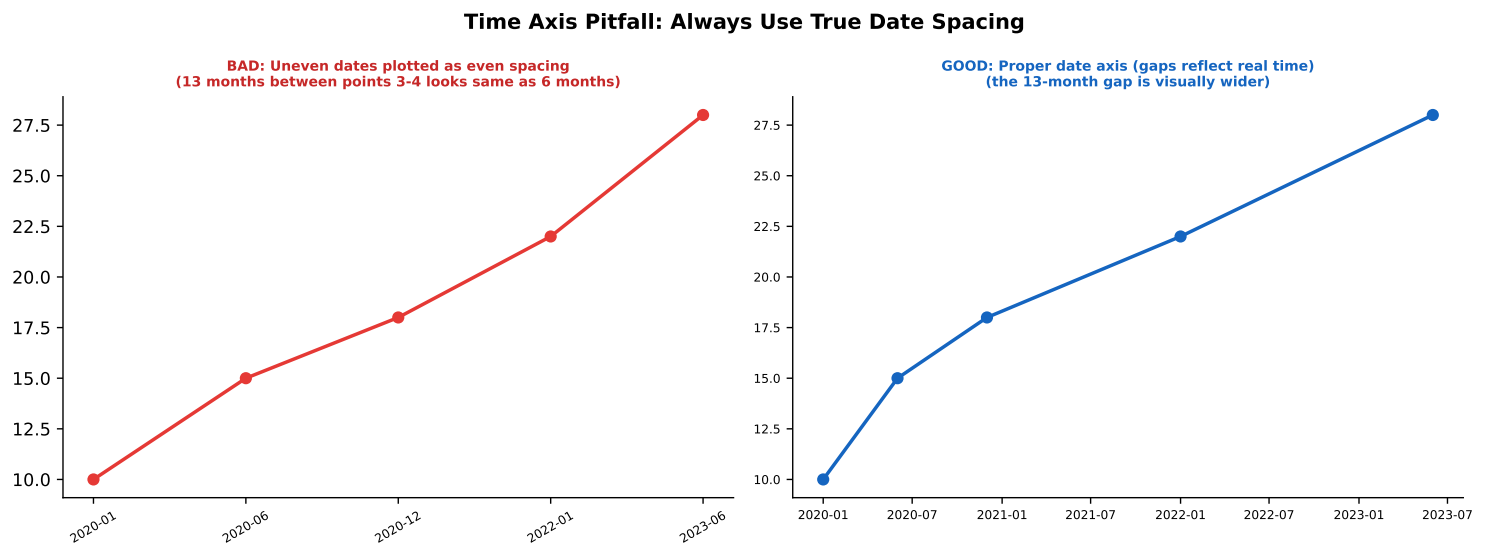


Choosing the Right Chart

Question	Chart	Why
How does Y change over time?	Line	Continuous trajectory
What is the cumulative total?	Stacked area	Parts sum to whole
When did Y change discretely?	Step	Rate changes, policies
What are the period totals?	Bar (temporal)	Monthly/yearly sums
What is the daily pattern?	Heatmap calendar	Day-level grid
How do many series compare?	Sparklines	Compact, Tufte-style

Pitfalls

Time-Axis Pitfalls



Pro-Tip

Three more pitfalls: (1) **Truncated y-axis** — starting at a high value exaggerates small changes. (2) **Missing dates** — gaps in the data should appear as gaps in the line, not as connected straight segments. (3) **Ignoring time zones** — always record and document the time zone.

Synthesis

Key Takeaways

1. Time is **ordered, continuous, hierarchical, and causal** — demanding specialised chart types and axis formatting.
2. **Granularity**: start fine (daily), aggregate up. Daily for events, weekly for medium spans, monthly for seasonal analysis, yearly for long trends.
3. **Decomposition**: $Y = T + S + R$ (additive) or $Y = T \times S \times R$ (multiplicative). Use STL for robust decomposition. R: `stl()`. Python: `statsmodels.tsa.seasonal.STL()`.
4. Five temporal chart types: **line** (trend), **area** (cumulative), **step** (discrete changes), **bar** (period totals), **heatmap calendar** (daily patterns).
5. **Never** use categorical spacing for dates. Always use a true date/time axis.

Essential Readings

- Cleveland, R. B. et al. (1990). "STL: A Seasonal-Trend Decomposition Procedure Based on Loess." *Journal of Official Statistics*, 6(1).
- Hyndman, R. J. & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*, 3rd ed. OTexts. `https://otexts.com/fpp3/`
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 13 (Visualizing Time Series).

Next Module: Line Charts — Theory & Best Practices (W06-M02)